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Does reef architectural complexity influence resource availability for a large reef-dwelling invertebrate?

Enrique Lozano-Álvarez^{a*}, Nelia Luviano-Aparicio^a, Fernando Negrete-Soto^a, Cecilia Barradas-Ortiz^a, Sergio Aguíñiga-García^b, Piedad S. Morillo-Velarde^{a,c}, Lorenzo Álvarez-Filip^a, Patricia Briones-Fourzán^a

^aUnidad Académica de Sistemas Arrecifales, Instituto de Ciencias del Mar y Limnología, Universidad Nacional Autónoma de México. Prol. Av. Niños Héroes s/n, Puerto Morelos, Quintana Roo, 77580 México.

^bLaboratorio de Espectrometría de Masas, Centro Interdisciplinario de Ciencias Marinas, Instituto Politécnico Nacional. Apdo. Postal 592, La Paz, Baja California Sur, 23096 México.

^cPresent address: CONACYT– Instituto de Ciencias Marinas y Pesquerías, Universidad Veracruzana, Hidalgo 617, Col. Río Jamapa, Boca del Río, Veracruz, 94290 México.

*Corresponding author:

Enrique Lozano-Álvarez, elozano@cmarl.unam.mx, elozano.lozano6@gmail.com

Abstract

In coral reefs, loss of architectural complexity and its associated habitat degradation is expected to affect reef specialists in particular due to changes in resource. We explored whether these features could potentially affect populations of a large invertebrate, the spotted spiny lobster *Panulirus guttatus*, which is an obligate Caribbean coral reef-dweller with a limited home range. We selected two separate large coral reef patches in Puerto Morelos (Mexico) that differed significantly in structural complexity and level of degradation, as assessed via the rugosity index, habitat assessment score, and percent cover of various benthic components. On each reef, we estimated density of *P. guttatus* and sampled lobsters to analyze their stomach contents, three different condition indices, and stable isotopes ($\delta^{15}\text{N}$ and $\delta^{13}\text{C}$) in muscle. Lobster density did not vary with reef, suggesting that available crevices in the less complex patch still provided adequate refuge to these lobsters. Lobsters consumed many food types, dominated by mollusks and crustaceans, but proportionally more crustaceans (herbivore crabs) in the less complex patch, which had more calcareous macroalgae and algal turf. Lobsters from both reefs had a similar condition (all three indices) and mean $\delta^{15}\text{N}$, suggesting a similar quality of diet between reefs related to their opportunistic feeding, but differed in mean $\delta^{13}\text{C}$ values, reflecting the different carbon sources between reefs and providing indirect evidence of individuals of *P. guttatus* foraging exclusively over their home reef. Overall, we found no apparent effects of architectural complexity, at least to the degree observed in our less complex patch, on density, condition, or trophic level of *P. guttatus*.

Key words: Coral reef degradation; spotted spiny lobster; *Panulirus guttatus*; carbon and nitrogen stable isotopes; stomach contents; Caribbean Sea

1. Introduction

Habitat architecture can greatly influence the abundance and diversity by providing more niches and ways of exploiting environmental resources (Bruno and Bertness, 2001). In Caribbean coral reefs, architectural complexity depends on the presence of key reef-building, morphologically-complex corals such as *Acropora* and *Orbicella* species, which have suffered rapid geographical declines from diseases and other factors associated with anthropogenic activities, resulting in loss of architectural complexity (Álvarez-Filip et al., 2009, 2011). In addition to coral decline, habitat degradation is evident in reefs with local increases of macroalgae (Bruno et al., 2009, Nelson et al., 2016, Suchley et al., 2016).

Habitat degradation is expected to result in loss of species richness and abundance of reef fishes, particularly reef specialists (i.e., fishes that live exclusively in coral reefs) (Munday, 2004; Álvarez-Filip et al., 2011, 2015). This would be consistent with a greater vulnerability of specialists to disturbances (e.g. McKinney, 1997). However, predictions for invertebrate taxa are less clear, as different studies have obtained contrasting results (see Graham and Nash, 2013). For example, in Indonesia, Fuchs (2013) observed that more complex reef habitats had greater abundance and diversity of echinoderms, ascidians and mollusks, and Fabrizius et al. (2014) found that loss of architectural complexity was associated with a decrease in many macroinvertebrate groups, especially predation-prone mobile taxa, including crustaceans. By contrast, Idjadi and Edmunds (2006) reported a positive relation between coral traits and diversity, but not abundance, of reef-associated invertebrates, with topographic complexity accounting for a great proportion of the variation in diversity, whereas Nelson et al. (2016) found that dead coral habitats support invertebrate assemblages that can be more diverse and abundant than live coral habitats, underlining the important role of structural complexity in the biodiversity enhancing function of coral reefs. These studies suggest that irrespective of level of degradation

(as measured by percent cover of macroalgae versus live coral), reefs with dead coral may maintain a sufficiently complex 3-D framework so as to harbor a diverse invertebrate community.

Panulirus guttatus (Latreille, 1804) (Crustacea: Decapoda: Achelata: Palinuridae) is a relatively small species of spiny lobster that occurs throughout the Wider Caribbean region and is an obligate reef-dweller (Sharp et al., 1997), i.e., a habitat specialist (Briones-Fourzán and Lozano-Álvarez, 2013). After a protracted larval phase with multiple planktonic stages that develop in oceanic waters, the nektonic postlarvae of *P. guttatus* swim back to coastal areas and settle directly in the coral reef habitat, where these lobsters remain for the rest of their life. Subadults and adults of *P. guttatus* have a home range of ~100 m in radius and do not venture far from the reef structure, especially avoiding sandy areas (Lozano-Álvarez et al. 2002, 2007; Robertson and Butler, 2009). Therefore, individuals of *P. guttatus* that settle in an isolated reef or reef patch surrounded by sand possibly remain and forage opportunistically on that single reef for their entire benthic life (Negrete-Soto et al., 2002; Robertson and Butler, 2009). In addition, because *P. guttatus* are quite vulnerable to predators (Briones-Fourzán and Lozano-Álvarez, 2008), they depend for survival on the local availability of reef crevices deep enough for them to fully retreat into during the hours of light (Sharp et al., 1997; Lozano-Álvarez and Briones-Fourzán, 2001; Lozano-Álvarez et al., 2007).

Because resource specialization is a critical factor in determining a species vulnerability to disturbances, especially resource depletion (i.e., food and refuge) (McKinney, 1997), the aims of the present study were to explore the potential influence of reef complexity on population density and feeding ecology (nutritional condition, diet, and $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in tail muscle) of *P. guttatus*, a highly habitat-specialized species. We expected to find higher levels of lobster density and nutritional condition on a more complex, less degraded reef patch than on a less complex, more degraded reef patch.

2. Materials and Methods

2.1. Study area

The study was conducted in the Puerto Morelos Coral Reef National Park, on the Caribbean coast of Mexico. The Park includes an extended fringing coral reef composed of several elongated reefs and reef patches of various sizes and levels of architectural complexity, separated from the coast by a shallow (<5 m in depth) and narrow (<1500 m in width) reef lagoon. Towards the southernmost boundary of the Park lies a series of large reef patches called “Jardines” (centered at 20°50’01” N, 86°52’36” W), where *P. guttatus* is known to abound (Negrete-Soto et al., 2002; Lozano-Álvarez et al., 2007). Here, we chose two reef patches greatly differing in relief and apparent level of degradation. For convenience, the patches were called “Jardines H” (for “healthier”) and “Jardines D” (for “degraded”). Large massive *Orbicella* corals are the main reef-building corals in these two reefs, which were separated from each other by ~500 m and were surrounded by open sandy bottoms into which *P. guttatus* lobsters do not venture (Lozano-Álvarez et al., 2002; Robertson and Butler, 2009).

2.2. Assessment of architectural complexity and benthic community structure

The bathymetric profile of Jardines H and D was obtained by repeatedly sailing over them in a boat fitted with an echo sounder (Garmin EchoMap 50/70 Series) and a GPS (Garmin GPS MAP 500/700 Series). We measured the maximum length and width of each reef to roughly estimate its surface area. Using SCUBA, architectural complexity was estimated using two complementary methodologies: the rugosity index (RI), which is a measure of local relief (Risk, 1972), and the habitat assessment score (HAS), which is a larger-scale visual categorical estimate (Gratwicke and Speight, 2005). RI is the ratio of a straight line to the distance a flexible chain of equal length

travels when draped over the reef substrate (Risk, 1972). A flat surface would have an RI = 1, with larger numbers indicating more complex surfaces. Ten 25-m transects were randomly laid on each reef to characterize benthic communities and lobster density (see below), and a 25-m chain (1 cm link-length) was used to estimate reef rugosity along each transect (Álvarez-Filip et al., 2011). Multiplying the area of each reef by its RI provides an estimate of “reef volume”, which is a better estimate of the habitat available to crevice-dwellers such as *P. guttatus* (Robertson and Butler, 2009). HAS assigns a visual score from 1 (lowest) to 5 (highest) to each of six variables: rugosity at the reef-scape scale, variety of growth forms, habitat height, refuge size categories, percentage live cover, and percentage hard substratum (see Table I in Gratwicke and Speight, 2005). HAS is obtained by adding the scores of all six variables. A score of 6 (all variables with a score of 1) would represent the least complex habitat, whereas a score of 30 (all variables with a score of 5) would represent the most complex habitat. For each transect, we estimated HAS in 2 m × 2 m quadrats located at the beginning, midpoint, and end of each transect, and averaged the three scores as a measure of transect-level HAS. The data on RI and HAS were not normally distributed and were compared between reefs with Mann-Whitney *U*-tests. In addition, given that *P. guttatus* strongly depends on available shelter for survival, the “refuge size categories” data were further individually compared between reefs with a Student’s *t*-test. The categories of refuge size (in this case the largest entrance diameter of holes or crevices) are 1 (1–5 cm), 2 (6–15 cm), 3 (16–30 cm), 4 (31–50 cm), and 5 (>50 cm) (Roberts and Ormond, 1987; Gratwicke and Speight, 2005).

Benthic community structure was characterized on each reef using the point intercept method (Loya, 1972). The presence of live hard coral, dead coral, soft corals, sponges, other sessile invertebrates (e.g., anemones, *Millepora*, *Palythoa*), fleshy macroalgae, calcareous macroalgae, coralline algae, algal turf, bare hard substrate, and sand, was recorded every 25 cm

along the 25 m transect, resulting in 100 estimates per transect. The data on the benthic community structure were transformed to square root arc-sine and subjected to a principal component analysis.

2.3. *Lobster traits*

2.3.1. Estimates of density and lobster collection

Lobster density was estimated via nocturnal surveys using SCUBA dives. Using lamps, two divers thoroughly searched and quantified exposed and sheltered lobsters within 2 m to the left and 2 m to the right of the 25-m line transects (i.e., 10 belt transects per reef, each 25 m in length $\times$ 4 m in width = 100 m²). Density (number of lobsters per 100 m²) was compared between reefs with a Mann-Whitney test.

After estimating lobster density, we collected lobsters from each reef using hand nets, also at night, to increase the probability of finding contents in their stomachs. Within 1 h of collection, the lobsters were taken to the laboratory in ice slurry to stop digestion and were sexed, measured with Vernier calipers (carapace length, CL, from between the rostral horns to the posterior edge of the carapace, $\pm$ 0.1 mm), and weighed (W, in g) on a digital scale after blotting excess water. Lobsters were then dissected to obtain samples of tail (abdomen) muscle and to extract their stomach and hepatopancreas (digestive gland). The hepatopancreas was blotted and weighed (HW, $\pm$ 0.01 g). The stomach was preserved in 70% ethanol for further analysis.

2.3.2. Estimates of nutritional condition

We estimated three indices of nutritional condition for each lobster: the W/CL ratio, the hepatosomatic index ($HSI = HW/W \times 100$), and the blood (hemolymph) refraction index (BRI). In spiny lobsters, the W/CL ratio and the HSI are positively related with nutritional condition

(Robertson and Butler, 2000; Briones-Fourzán et al., 2009), although the W/CL ratio needs to be compared between lobsters over a similar size range (Briones-Fourzán et al., 2009). Because hemolymph density is positively correlated with protein concentration in the hemolymph, BRI is a proxy for physiological condition once molt stage is taken into consideration (Lorenzon et al., 2011; Rodríguez-García et al., 2015). We extracted hemolymph from the base of the fifth pereopod of each lobster using an insulin syringe and read it in a density refractometer. The tip of a pleopod was excised for observation under a microscope to determine molt stage (see Lyle and MacDonald, 1983). Only lobsters in intermolt (stage C of Drach's scale) were considered for estimates of BRI. The three nutritional indices of lobsters were compared between reefs with Student's t-tests.

2.3.3. Stomach content analyses

The percent fullness of the stomach was visually estimated and stomach contents were sorted under a stereomicroscope. Although some items were identified to species, most were identified to higher levels due to their fragmentation and/or partial digestion. Contents were grouped for data analysis in food categories corresponding to gross taxa. For each food category, we estimated percent volume (%V = percent contribution to the total volume of stomach contents), percent frequency of occurrence (%F = number of stomachs containing the food category/total number of stomachs examined $\times$ 100), and an index of relative importance [IRI = (%V $\times$ %F)/100] (Briones-Fourzán et al., 2003, Segura-García et al., 2016). IRI values range from 0 to 100, with values >40 denoting preferred prey, values between 10 and 40 denoting secondary prey, and values <10 denoting occasional or accidental prey (Pinkas et al., 1971). Using the %V of food categories, we computed Horn's index of overlap (Horn, 1966) as a measure of diet overlap between lobsters from both reefs. Horn's index ranges from 0 (no overlap) to 1 (complete

overlap). For conspecific groups, index values ≤ 0.8 are considered to be indicative of major differences in diet (Cartes and Sardà, 1989; Briones-Fourzán et al., 2003).

2.3.4. Stable Isotope Analyses (SIA)

For SIA, samples of abdominal (tail) muscle from 20 lobsters from each reef were rinsed with ultrapure MilliQ water, dried completely at 60 °C, and ground to a fine powder with agate mortar and pestle. Approximately 1 mg of fine powdered tissue was transferred into tin capsules. For all samples, the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ composition was determined at the Mass Spectrometry Laboratory, Centro Interdisciplinario de Ciencias Marinas, La Paz, Mexico. The determination was conducted using an isotope ratio mass spectrometer (Finnigan Delta V Plus, Thermo Scientific) coupled to an elemental analyzer (Elemental Combustion System, Costech model 4010) with an average precision across runs of 0.02‰ for $\delta^{13}\text{C}$ and 0.1‰ for $\delta^{15}\text{N}$. Carbon and nitrogen ratios were expressed in delta (δ) notation, in units of per mil (‰): $\delta = 1000 \times (R_{\text{sample}} - R_{\text{standard}})/R_{\text{standard}}$, where R is the ratio of the heavier, rarer isotopes (^{13}C and ^{15}N) to the lighter, more common isotopes (^{12}C and ^{14}N , respectively) (Post, 2002). Delta values are reported relative to the international standards of Vienna Pee-Dee Belemnite carbon and atmospheric nitrogen. The isotopic niche width for lobsters from each reef was determined based on the isotopic dispersion of samples within a two-dimensional ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) space. We used two separate ANCOVAs to assess the effects of lobster size (covariate) and reef (categorical variable) on values of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$.

3. Results

3.1. Architectural complexity and benthic community structure

Both study reefs differed in most features. Jardines H had a surface area of 11,000 m² and an

abrupt relief formed by the skeletons of massive hard corals, whereas Jardines D had a surface area of 14,100 m² and a much flatter relief (Fig. 1).

The two reefs varied significantly in RI (median [interquartile range], Jardines H: 1.91 [1.31 to 2.44], Jardines D: 1.09 [1.05 to 1.14], $U = 8.0$, $n_1 = n_2 = 10$, $Z = 3.137$, $p = 0.002$) as well as in HAS (Jardines H: 15.0 [14.4 to 16.3], Jardines D: 12.2 [10.8 to 12.8], $U = 21.5$, $n_1 = n_2 = 10$, $Z = 2.117$, $p = 0.034$), with Jardines H exhibiting a higher degree of architectural complexity. This was also reflected in a greater reef volume (area $\times$ average rugosity) for Jardines H (21,392) than for Jardines D (15,260). Refuge size categories ranged from 1 (1–5 cm) to 3 (15–30 cm) on Jardines H, whereas all refuge size categories over Jardines D were either 1 or 2 (5–10 cm). Therefore, available refuges were larger in Jardines H (1.7 ± 0.3 , mean size category $\pm$ 95% CI) than in Jardines D (1.13 ± 0.2) ($t_{18} = 3.08$, $p = 0.006$).

Despite great within-reef variability, the dominant benthic components on Jardines H were fleshy macroalgae, soft corals, and dead coral, whereas Jardines D was dominated by algal turf, calcareous macroalgae, and fleshy macroalgae (Table 1). Several individual benthic components varied significantly with reef, with proportionally more live hard coral, soft corals, and other sessile invertebrates (e.g., *Millepora*, *Palythoa*, anemones) on Jardines H, and proportionally more algal turf and calcareous macroalgae on Jardines D (Table 1). Results of the PCA revealed that the first two components explained 69.2% of the total variance. The first component explained 47.9% of the variance and was positively correlated with dead coral and negatively correlated with turf and calcareous macroalgae. The second component explained 21.3% of the variance and was strongly positively correlated with sand and negatively correlated with reef calcifying organisms: coralline algae and live hard coral (Fig. 2). Most transects on Jardines H differed from those on Jardines D along the second component.

3.2. Lobster traits

Lobster density ranged from 0 to 6 lobsters 100 m^{-2} on Jardines H (median [interquartile range], 3.0 lobsters [2.2 to 3.8] 100 m^{-2}) and from 0 to 12 lobsters 100 m^{-2} on Jardines D (3.0 lobsters [2.0 to 5.8] 100 m^{-2}), with no significant effect of reef ($U = 46.0$, $n_1 = n_2 = 10$, $Z = 0.265$, $p = 0.791$).

Nutritional indices and stomach content analyses were conducted on 60 lobsters (29 from Jardines H, 31 from Jardines D) over a size range of 29.1 to 72.0 mm CL. The mean size of lobsters was slightly, but not significantly, larger in Jardines H (mean $\pm$ SD: 55.9 ± 2.8 mm CL) than in Jardines D (51.3 ± 3.6 mm CL) ($t_{58} = 1.957$, $p = 0.056$). W/CL ratios and HSI were estimated in all 60 lobsters, but BRI was estimated in only 37 lobsters that were in intermolt (18 and 19 from Jardines H and D, respectively). None of the three indices of nutritional condition of lobsters differed significantly between reefs (Table 2).

Three lobsters from Jardines H and four from Jardines D had empty stomachs. Lobsters from both reefs fed primarily on mollusks (mostly gastropods and some bivalves), crustaceans (mostly majoid crabs and some hermit crabs), and echinoderms (sea urchins, brittle stars), but also on polychaetes and sponges, with macroalgae present in about 30% stomachs of lobsters from either reef (Table 3). In terms of gross taxa, the diet of lobsters from both reefs exhibited significant overlap (Horn's index: 0.917), but no food category reached an IRI of 40. That is, no food category can be considered as preferred, reflecting the omnivore habits and opportunistic feeding of *P. guttatus*. In terms of volume, mollusks were the most important component in the diet of lobsters, but with differences between reefs. Mollusks contributed 40% to the volume of stomach contents in lobsters from Jardines H, but only 30.7% in lobsters from Jardines D. Conversely, crustaceans contributed proportionally more in volume to the diet of lobsters on Jardines D (26.5%) than on Jardines H (6.9%). This pattern was reflected in a higher IRI of

crustaceans for lobsters in Jardines D (12.8) than in Jardines H (1.9) (Table 3).

The isotopic dispersion of lobsters within the two-dimensional ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) space showed a clear separation of the lobster samples from the two reefs along the $\delta^{13}\text{C}$ axis, but not along the $\delta^{15}\text{N}$ axis (Fig. 3), suggesting differences in carbon sources between reefs. Results of ANCOVAs showed that values of $\delta^{13}\text{C}$ in muscle tissue tended to increase with lobster size ($p = 0.05$) but varied significantly with reef (Table 4, Fig. 4a), with reef explaining 63% of the variation in the values of $\delta^{13}\text{C}$ versus 4% explained by size. In contrast, values of $\delta^{15}\text{N}$ increased significantly with lobster size but did not vary with reef (Table 4, Fig. 4b), with lobster size explaining about 20% of the variation in of $\delta^{15}\text{N}$ values.

4. Discussion

In Caribbean coral reefs, a greater abundance of algae is usually associated with coral death and habitat degradation (Nelson et al., 2016; Suchley et al., 2016). Because the main reef-building corals in our study sites are large massive *Orbicella* corals, it is likely that the major drivers of reef degradation of these reef sites are biological disturbances (e.g., climate-induced coral bleaching and coral disease) that kill coral tissue without immediately compromising the integrity of coral skeletons. Eventually, gradual declines in reef architectural complexity could occur if the carbonate budget of the reefs became negative due to the loss of reef-building corals and the increase of biological and physical erosion (Pratchett et al., 2008).

The aim of this study was to examine the potential effects of reef architectural complexity and associated level of degradation on several traits of an obligate coral-reef dweller, *Panulirus guttatus*, which might reflect the level of availability of resources (shelter and food). While our findings based on the comparison of two small reefs may not be representative of the impact of reef degradation on *P. guttatus* populations at large, these reefs did represent two significantly

different levels of habitat complexity and degradation, hence providing a basis on which further investigations can be conducted. Indeed, our results showed that Jardines H was more complex and less degraded, with higher values of rugosity index (RI) and habitat assessment score (HAS), less cover of algal turf and calcareous algae, and more cover of live hard coral and soft corals. In contrast, Jardines D was less complex and more degraded, with lower values of RI and HAS, greater expanses of turf and calcareous macroalgae, and virtually no live coral cover.

Yet, many of the traits of *P. guttatus* that we examined did not vary with reef despite the difference in their architectural complexity and level of degradation. For example, the greater complexity, “reef volume”, and mean size of refuges on Jardines H were not reflected in a greater lobster density in that reef than in Jardines D. Although we only measured density once, prior to collection of lobsters for dissection, the median lobster density for both reefs (3 lobsters 100 m^{-2} , or 300 per hectare) was consistent with other visually-estimated densities of *P. guttatus* at Puerto Morelos reefs, as reported by Lozano-Álvarez et al. (2007) (average of three reefs: 287 lobsters ha^{-1} on the fore reef zone) and Briones-Fourzán et al. (2013) (3.1 and 4.9 lobsters 100 m^{-2} , or 310 and 490 lobsters ha^{-1} , on two reefs separated by 11 km). Wynne and Côté (2007) visually estimated density of *P. guttatus* at 9.4 to 234.2 lobsters ha^{-1} over a range of fished and unfished reefs in Anguilla, with significantly more lobsters in “high quality” reefs (defined as reefs with more crevice shelters and less macroalgae, sand, and sediment cover) than in “low quality” reefs, irrespective of whether the reefs were fished or not fished for lobsters. In contrast, in Florida (USA), Robertson and Butler (2009) visually estimated density of *P. guttatus* at 0.0083 to 0.25 lobsters per 100 m^2 (or 0.8 to 25 lobsters ha^{-1}) on degraded patch reefs of smaller sizes (152 m^2 to 8000 m^2) than Jardines H and D. They found that lobster abundance was greater on patch reefs of greater area and volume but lobster density was significantly higher at smaller reef sizes and volumes.

A possible explanation for these apparently counterintuitive findings is that population density of *P. guttatus* lobsters is more affected by the availability of appropriately scaled shelters offering protection from predation (Sharp et al., 1997; Lozano-Álvarez and Briones-Fourzán, 2001), rather than by the size or volume of their home reefs. For example, the smallest cavity in which Wynne and Côté (2007) observed *P. guttatus* measured $5 \times 10 \times 25$ cm depth, i.e., had an entrance size corresponding to categories 1 and 2 of the “refuge size” variable in HAS, which were the more abundant categories in both reefs. Thus, it is possible that many of these refuges were of appropriate dimensions for individuals of *P. guttatus*.

Although lobsters from the two reefs exhibited a high degree of diet overlap in terms of gross taxa, their diet differed in the proportional volumetric contribution of the two most important food categories: mollusks and crustaceans. Mollusks had been previously found to be the most important prey in stomachs of *P. guttatus* from Puerto Morelos, followed by crustaceans (Colinas-Sánchez and Briones-Fourzán, 1990; Segura-García et al., 2016). However, our samplings conducted in separate, discrete reefs, revealed that lobsters from Jardines D consumed proportionally more crustaceans and fewer mollusks than lobsters from Jardines H. Most crustacean prey consisted of small herbivorous crabs of the genera *Mithraculus*, *Mithrax*, and *Teleophrys*, which had been previously reported as prey for *P. guttatus* (Colinas-Sánchez and Briones-Fourzán, 1990; Butler and Kintzing, 2016). It is known that variation in the proportion of different habitats (e.g. seagrass, macroalgae, sand, bare hard substrate) at the landscape scale (10s to 100s of meters) can influence the diet of spiny lobsters (e.g. *P. cygnus*: Edgar, 1990; MacArthur et al., 2011; *P. argus*: Cox et al., 1997, Briones-Fourzán et al., 2003, *Jasus edwardsii*: Jack and Wing, 2011). For example, total macroalgae, but particularly calcareous macroalgae and algal turf, were significantly more abundant on Jardines D than on Jardines H. At the regional scale, a greater algal cover on coral reefs does not necessarily correlate with an increase in

abundance of herbivore fishes (Suchley et al., 2016), but a recent study in other Puerto Morelos reefs found that, at the landscape scale, algal cover did correlate with a greater abundance of herbivore crabs such as *Mithraculus coryphe*, *Teleophrys ruber* (= *M. ruber*), and *M. sculptus* (González-Gómez, 2016). This finding may explain both the higher volumetric contribution of majoid crabs to the diet of *P. guttatus* and the isotopic enrichment in the heavier carbon isotope ^{13}C (values closer to zero) in lobsters from Jardines D. Butler and Kintzing (2016) also reported a high level of predation on herbivore crabs (mostly juvenile *Damithrax spinosissimus*) by *P. guttatus* in degraded reefs in Florida. *Damithrax*, *Mithrax*, and *Mithraculus* crabs can consume fleshy and calcareous macroalgae as well as epiphytes and turf (Coen, 1988a, 1988b; Stachowicz and Hay, 1996; Butler and Mojica, 2012). Because spiny lobsters consume many types of invertebrates and feed opportunistically on available prey, herbivore invertebrates (e.g. majoid crabs) may contribute disproportionately to the diet of *P. guttatus* in degraded reefs (Butler and Kintzing, 2016).

Although indices of nutritional condition of spiny lobsters can be affected by water temperature, molt stage, ontogenetic phase, degree of starvation, quality of food, and site (Oliver and McDiarmid, 2001; Behringer and Butler, 2006; Briones-Fourzán et al., 2009; Lopeztegui et al., 2012; Rodríguez-García et al., 2015), none of the three indices of nutritional condition of *P. guttatus* that we estimated differed between reefs. This result suggests that despite differences in the relative proportion of food categories and in carbon sources between Jardines H and D, the quality of the lobster diet was similar on both reefs. However, the separation of the lobster samples from both reefs along the $\delta^{13}\text{C}$ axis provides indirect evidence that foraging activity of individuals of *P. guttatus* is indeed constrained to the reef on which they reside. This constraint may have selected for a broad diet based on available prey.

The isotopic signature and trophic position of individuals may vary spatially or with size

in some species (e.g., Deudero et al., 2004; Waddington et al., 2008; Jack and Wing, 2011) but not in others, although small sample sizes and spatial variability may mask such relationships (Deudero et al., 2004). For example, Guest et al. (2009) found no apparent relationship between values of $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ and size in red rock lobsters *Jasus edwardsii* collected from several sites in Tasmania, Australia, suggesting that lobsters over the size range examined fed on similar food types. This was also the case for 14 *P. guttatus* collected by Segura-García et al. (2016) from multiple sites along the Puerto Morelos coral reef. We found a positive relationship between values of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and lobster size, which might suggest changes in the diet of these lobsters as they grow, i.e., that larger lobsters obtain a greater proportion of their diet from higher trophic levels. However, this issue requires further investigation because this result could also reflect metabolic age effects not necessarily related with diet (e.g. Overman and Parrish, 2001).

Nelson et al. (2016) considered coral reefs with a live coral cover of 20–40% as “healthy” and reefs with a live coral cover <5% as “degraded”. Based on these values, Jardines D could be considered as highly degraded and Jardines H as close to being degraded, yet the latter was significantly more complex than the former. Nevertheless, our results suggest that the differences in complexity and degradation between these reefs are not sufficiently broad so as to affect the density or feeding ecology of their local populations of *P. guttatus*. However, loss of architectural complexity and degradation of Caribbean coral reefs is an ongoing process that may further increase due to multiple stressors (Bruno et al., 2009; Álvarez-Filip et al., 2009). Studies on reef fishes predict that this process will affect reef specialists in particular (Graham et al., 2007; Álvarez-Filip et al., 2011) because these fishes tend to favor complex corals that have branching or plating functional forms (Munday, 2004). This does not appear to be the case for *P. guttatus*, as these lobsters can use holes and crevices in the reef 3-D framework irrespective of whether the coral is alive or dead, and their omnivory and opportunistic feeding allows them to feed on

various types of available prey. Other obligate coral-dwelling organisms have also been found to dwell equally well on dead as on live coral, suggesting that habitat associations are more complex than they would appear (Head et al., 2015). However, habitat availability for *P. guttatus* might be dependent on long-term coral loss, given that degraded coral reef habitats further erode over time (Perry et al., 2012). Therefore, more work is required to establish the vulnerability of *P. guttatus* and other specialist reef invertebrates to flattening and degradation of coral reefs.

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Figure legends

Fig. 1. Location (left panels) and bathymetric profile of the studied reefs, Jardines H (right upper panel) and Jardines D (right lower panel). Degrees are in decimal notation.

Fig. 2. Principal Components Analysis (PCA) bi-plot on arcsine square-root transformation of percent cover of 11 benthic components over the two studied reefs, Jardines H (grey dots) and Jardines D (white dots). Each dot represents a transect. LC: live hard coral, DC: dead coral, BS: bare hard substrate, AT: algal turf, SA: sand, SC: soft corals, SP: sponges, FM: fleshy macroalgae, CM: calcareous macroalgae, CA: coralline algae, OT: other sessile invertebrates.

Fig. 3. Bi-plot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ isotopic values of spiny lobsters *Panulirus guttatus* from two reefs differing in architectural complexity and degradation, Jardines H (grey dots) and Jardines D (white dots). Error bars denote 1 SD.

Fig. 4. Relationship between carapace length (mm) and values of $\delta^{13}\text{C}$ (upper panel) and $\delta^{15}\text{N}$ (lower panel) of *Panulirus guttatus* lobsters collected on two separate reefs, Jardines H (grey dots) and Jardines D (white dots).

Fig. 1

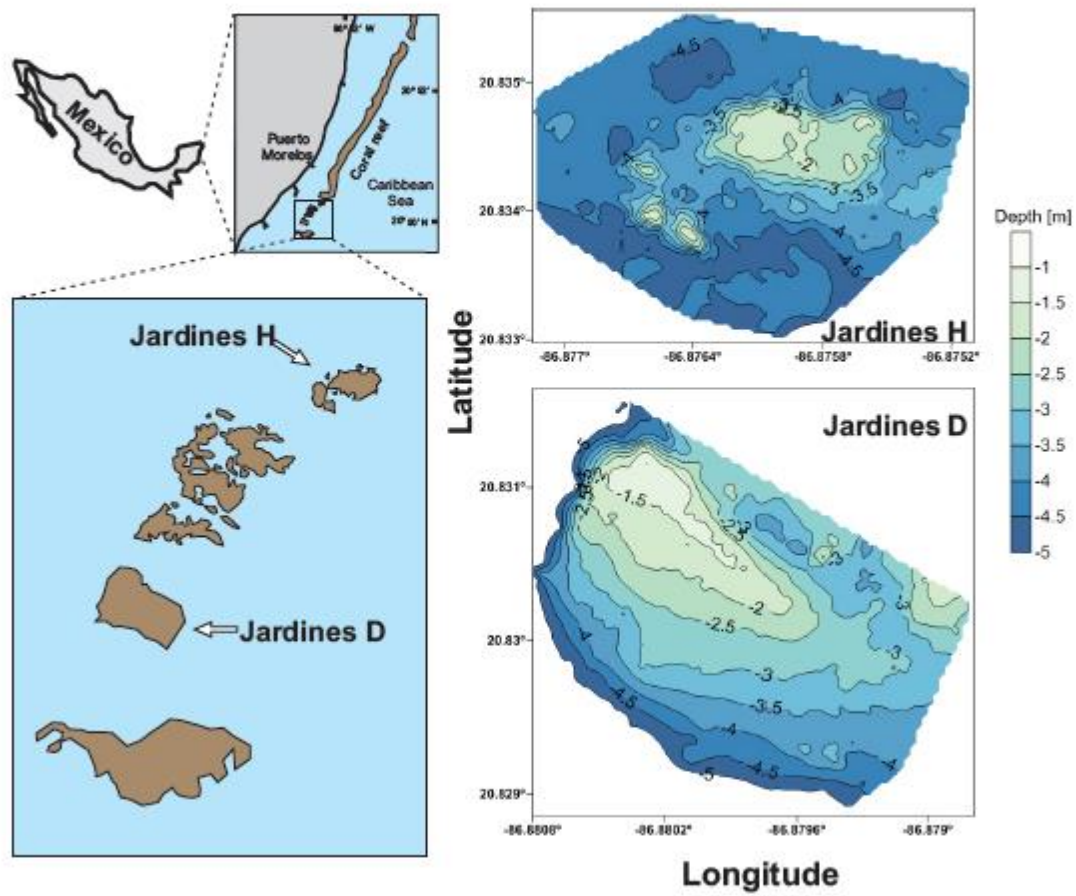


Fig. 2

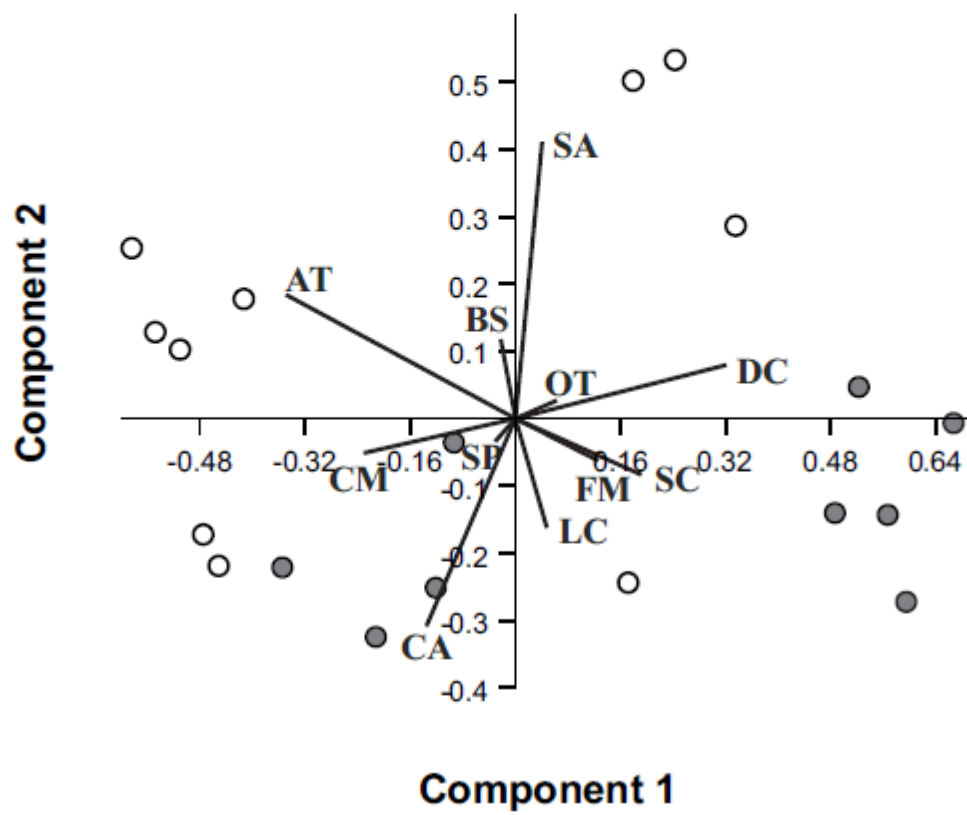


Fig. 3

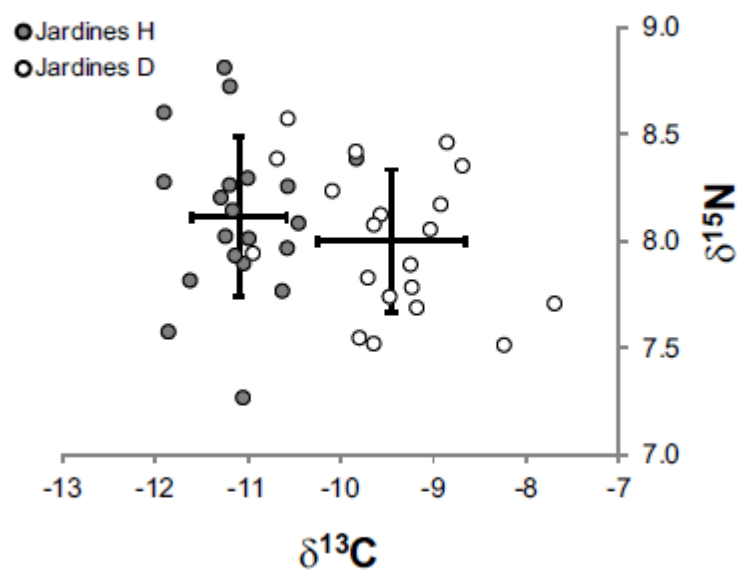


Fig. 4

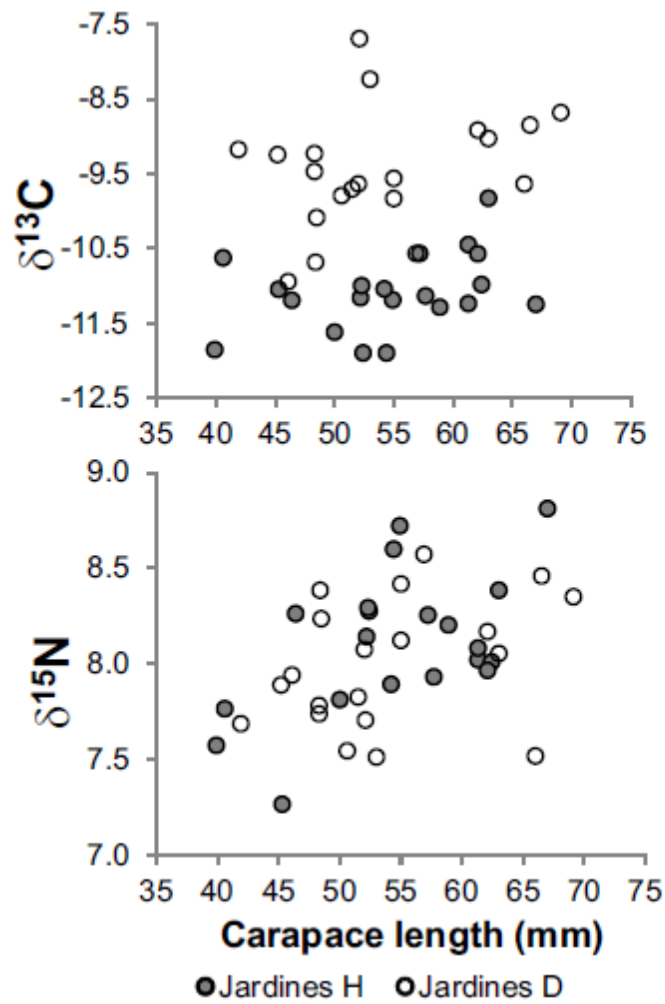


Table 1. Percent cover (mean $\pm$ SE) of benthic components on the studied reefs as measured in 10 transects per reef. Total algal cover is the sum of the four last components. Student's *t*-tests were done on arcsine-square root transformed data. Degrees of freedom = 18 in all cases.

Benthic component	Jardines H	Jardines D	<i>t</i>	<i>p</i>
Live hard coral	7.9 $\pm$ 2.1	1.0 $\pm$ 0.4	3.089	0.006
Dead coral	15.2 $\pm$ 4.4	6.9 $\pm$ 3.1	1.441	0.167
Soft coral	17.2 $\pm$ 3.1	4.3 $\pm$ 1.5	3.951	0.001
Sponges	5.0 $\pm$ 1.5	5.2 $\pm$ 1.1	-0.341	0.737
Other sessile invertebrates	3.4 $\pm$ 1.0	0.8 $\pm$ 0.5	2.147	0.046
Bare hard substrate	0.6 $\pm$ 0.5	1.9 $\pm$ 0.7	-1.753	0.097
Sand	7.2 $\pm$ 5.4	10.6 $\pm$ 4.6	-0.585	0.566
Total algal cover	43.5 $\pm$ 6.6	69.3 $\pm$ 8.1	-2.559	0.020
Algal turf	5.2 $\pm$ 2.2	24.4 $\pm$ 7.3	-2.570	0.019
Fleshy macroalgae	19.6 $\pm$ 4.1	15.8 $\pm$ 2.2	0.710	0.487
Calcareous macroalgae	10.6 $\pm$ 4.1	21.3 $\pm$ 3.0	-2.555	0.020
Coralline algae	8.1 $\pm$ 3.3	7.8 $\pm$ 3.5	0.357	0.725

Table 2. Nutritional indices estimated for *Panulirus guttatus* lobsters from each reef (Jardines H and D). W = lobster weight, CL = carapace length.

Nutritional index	Jardines H	n	Jardines D	n	t	df	p
W/CL ratio	2.96 ± 0.24	29	2.61 ± 0.35	31	1.597	58	0.116
Hepatosomatic Index	4.14 ± 0.21	29	3.93 ± 0.30	31	1.135	58	0.261
Blood Refractive Index	1.365 ± 0.002	18	1.357 ± 0.003	19	1.350	35	0.186

Table 3. Summary of stomach content analyses for *Panulirus guttatus* lobsters collected from Jardines H (n = 29) and Jardines D (n = 31). %V: percent volume; %F: percent frequency; IRI: Index of Relative Importance = $(\%F \times \%V)/100$, Others: unidentified remains and organic matter.

Food categories	Jardines H			Jardines D		
	%V	%F	IRI	%V	%F	IRI
Mollusks	40.0	73.1	29.2	30.7	70.4	21.6
Crustaceans	6.9	26.9	1.9	26.5	48.1	12.8
Echinoderms	11.3	34.6	3.9	10.7	33.3	3.6
Polychaetes	7.5	23.1	1.7	11.7	18.5	2.2
Sponges	7.7	19.2	1.5	5.6	11.1	0.6
Macroalgae	18.7	34.6	6.5	10.0	29.6	3.0
Coralline algae	4.0	7.7	0.3	0	0	0
Others	3.8	15.4	0.6	4.8	7.4	0.4

Table 4. Results of separate ANCOVAs to test for effects of lobster size (carapace length, covariate) and reef (categorical predictor with 2 levels: Jardines H and Jardines D) on values of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in tail muscle of *Panulirus guttatus*.

	$\delta^{13}\text{C}$							$\delta^{15}\text{N}$					
	Sum of		Mean					Sum of		Mean			
Effect	squares	df	square	F	p	R ²		squares	df	square	F	p	R ²
Model	28.726	2	14.363	34.677	<0.001	0.67		2.73	2	1.365	12.694	<0.001	0.23
Intercept	101.433	1	101.43	244.89	<0.001			35.104	1	35.104	341.89	<0.001	
Size	1.707	1	1.707	4.122	0.05	0.04		0.934	1	0.934	9.101	0.005	0.20
Reef	27.603	1	27.603	66.643	<0.001	0.63		0.099	1	0.099	0.967	0.332	0.03
Error	15.325	37	0.414					3.799	37	0.103			

Highlights

- We compared *P. guttatus* traits in two reefs differing in complexity and degradation
- Density, nutritional condition, and mean $\delta^{15}\text{N}$ did not differ with reef
- Diet components and mean $\delta^{13}\text{C}$ differed with reef reflecting different carbon sources
- *P. guttatus* diets have the potential to reflect the changing nature of their ecosystem.
- *P. guttatus* is able to adjust its diet to maintain population size and condition as reefs degrade.